

DevOps Assignment

Roadpass Digital

Overview

Complete the three exercises below. Once you have completed your submission, follow the [Submission Instructions](#). We will schedule a time to discuss your solution with you after we have reviewed internally.

Please complete the assignment and return within 3 business days. If more time is needed, please let us know so we can accommodate.

AWS VPC Exercise

Create a new terraform module for creating a staging VPC in AWS. The following requirements should be met:

- The network supernet of 172.16.0.0/16
- The VPC should provide an endpoint for AWS Systems Manager (SSM), internally.
- The VPC should provide an internal endpoint for S3.
- The VPC should have two different availability zones, with two private subnets and two public subnets for each availability zone.
- The VPC should contain appropriate routing tables.
- The VPC should include appropriate NAT gateways and an internet gateway.

Create module code to fulfill this request utilizing terragrunt/terraform. Show how you would use this module code to bring up the given architecture.

A Small EC2 App

Use the following technologies

- [Packer](#)
- [Ansible](#)
- Terraform/terragrunt.

Utilizing a pack/fry idiom in ansible, create an Amazon Machine Image (AMI) which runs a small nginx webserver. (A static page loading here is just fine).

- It should be based on the latest Amazon Linux 2023 AMI.
- Ensure there are both pack and fry roles and that the appropriate variables needed for the ami are injectable via userdata in the frying role. (if applicable)

Create a terraform stack which creates the following configuration:

- An autoscaling group of two Ec2 Instances
- Launch template
 - Instance profile role along with manageable policy to be attached to the EC2 instances.
- An application load balancer (ALB)
 - A target group where you will place the ec2 instances.
- Security groups for the load balancer and ec2 instances with the proper rules to ensure ssh access to the instances and communication between the instances and the load balancer.
- Place the autoscaling group on a private subnet.
- Bonus points if you add transport layer security support via an SSL cert through AWS Certificate Manager (ACM) but not necessary for this exercise.

Deploying An Application

Helm

- Create a simple templated helm chart that loads up an nginx server and an appropriate ingress. (assume either an nginx or alb controller is just fine)
- Create a values.yml file that fulfills the values for this helm chart.
- Show a command that dumps the template generated.

Github Actions

- Create an example github action that would use Github's Open ID Connect (OIDC) provider as a trusted AWS Identity to deploy the helm chart above into a staging EKS cluster.

Submission Instructions

- Create a GitHub repository (public or private) with all deliverables
- Include a main README.md with an overview of your submission
- Organize files cleanly
- Send the repository link to your hiring contact along with your actual time spent